Volatility Risk Premium and Non-Directional Option Strategies on the S&P 500 Index: Evidence from 2010–2023 SPX Option Chains

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Extended abstract

Question. This paper tests whether the prices of European options on the S&P 500 index contain a volatility risk premium (VRP) and whether non-directional premium-selling strategies (iron condors, credit spreads, strangles and related structures) capture it net of costs.

Data and method. End-of-day option chains of SPX options (European-style, cash-settled index options listed on Cboe; source OptionsDX) from January 2010 to December 2023: 3,499 sessions and 47,699 date–expiry pairs. Implied volatility is obtained from prices with the Black (1976) model on the forward implied by put-call parity; model-free implied variance follows the Cboe formula; realized variance is measured up to the expiry of the same option. The real-world forecast is a GJR-GARCH(1,1) with Student-t innovations estimated with no look-ahead. The backtests cover 434 configurations of ten families, for 1,310,509 simulated trades, at mid and at bid-ask prices.

Results. (i) The 30-day model-free volatility exceeds realized volatility by +3.45 points on average (t 6.48); at-the-money volatility exceeds it by +0.72 points (t 1.36). The premium sits in the wings, not at the center of the surface. (ii) The at-the-money premium is significant in 2010–2014 (+1.82 points, t 4.02) and zero in 2015–2019 and 2020–2023. (iii) The 30-day at-the-money implied volatility is an unbiased forecast of realized volatility (slope 0.908, joint test $p = 0.409$) with a smaller error than GARCH and historical volatility. (iv) 0.10-delta, 30-day puts returned to their buyers a fraction of 0.44 of the premium paid. (v) The median Sharpe ratio across configurations is +0.39 at mid prices and -0.40 at bid-ask prices; the share with a significantly positive alpha against the index falls from 16.4% to 2.5%, the value expected by chance alone. (vi) Positive results are concentrated in structures that sell puts and carry positive equity beta; iron condors and straddles have a zero or negative net result; early-exit rules lower returns because they pay a second spread. (vii) Suspending entries when realized volatility exceeds 25% does not improve the Sharpe ratio (median from -0.30 to -0.34) and does not reduce the worst month: extreme losses originate in calm phases, and delta-selected strikes already widen with implied volatility (from 3.8% to 7.5% of the forward between the low and the high tercile). (viii) Returns are far from normal: for the 0.16-delta, 45-day short put per-trade skewness is -9.5 and excess kurtosis 108 (index over the same horizon: -1.1 and 4.8), normality of the monthly series is rejected in 92% of configurations and the best configuration of each family is no exception (Figure 4: per-trade skewness from -6.7 for the short put to +7.0 for the long put). The Sharpe ratio therefore overstates short-volatility structures: tail risk has to be measured directly.

	Mid prices	Bid-ask prices
Median Sharpe ratio, 434 configurations	+0.39	-0.40
Share with positive Sharpe ratio (%)	64	38
Share with positive alpha, $t > 2$ (%)	16.4	2.5
Short put Δ0.16, 45 days: Sharpe	+0.99	+0.93
Put credit spread Δ0.20/0.05, 30 days: Sharpe	+0.98	+0.83
Iron condor Δ0.16/0.05, 30 days: Sharpe	+0.43	+0.14

Conclusion. The premium exists as compensation for tail risk and is consistent with a market that is informationally efficient at the center of the surface. The bid-ask spread, the embedded equity exposure and ex-post parameter selection explain almost all of the positive backtest results. The hypothesis that systematic capture of the premium tends, over the long run and net of costs, to a zero result is not rejected by the data.

Keywords: volatility risk premium, index options, iron condor, put credit spread, GJR-GARCH, transaction costs, multiple testing.

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Research document; not investment advice.

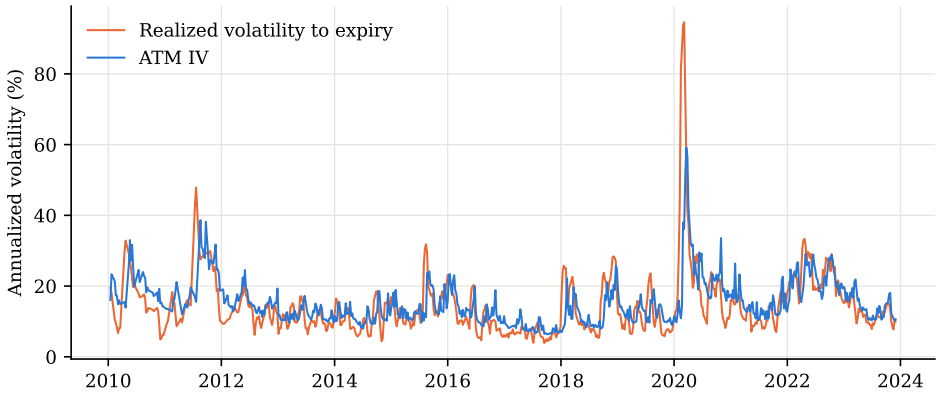


Figure 1. At-the-money IV on the expiry closest to 30 days and realized volatility to that expiry.

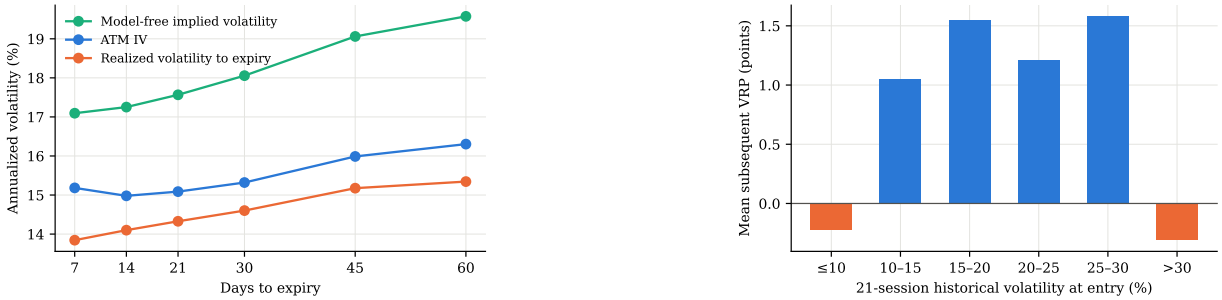


Figure 2. Left: means by horizon of model-free IV, at-the-money IV and realized volatility. Right: mean subsequent at-the-money premium by level of historical volatility at entry.

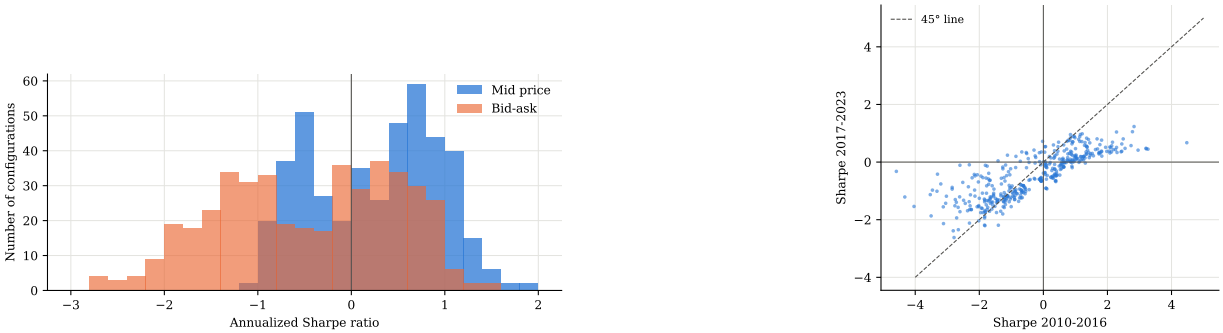


Figure 3. Left: distribution of the Sharpe ratio across the 434 configurations at mid and bid-ask prices. Right: Sharpe ratio in 2010–2016 and in 2017–2023 for credit configurations.

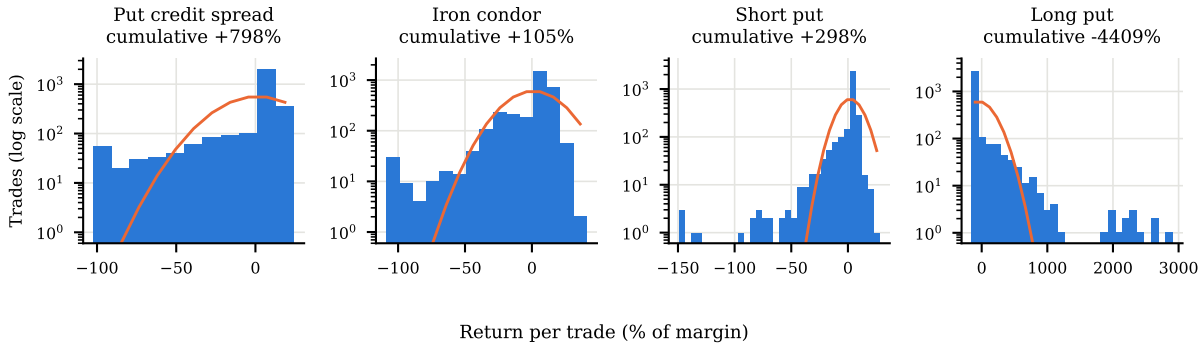


Figure 4. Return per trade on margin (bid-ask prices, logarithmic scale) of the configurations with the highest cumulative return for the put credit spread, the iron condor, the short put and the long put; the curve is the normal with the same mean and variance.